

Lecture 7

Lax pairs and isospectral flows

Objectives. *Lecture 4* said what integrability is and what it buys, and nothing about how to find the integrals. This lecture gives the method that finds them for the many-body models of Part I: write the equations of motion as a matrix whose spectrum does not change. We prove that the spectrum is conserved, derive such matrices for the oscillator and the open Toda chain, recognise one in Euler's equations for a rigid body, and say plainly what the method does not give — a debt that *Lecture 8* pays.

7.1 The gap L4 left

Definition 4.6 asks for n independent first integrals in involution, and *Theorem 4.8* turns them into tori. Neither says where the integrals come from (§ 4.6). The ones we own came from two sources: the energy, and symmetry — angular momentum for a central force, momentum for a translation invariant system.

That is not enough for the models this course is built around. The open Toda chain of N particles,

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}}, \quad (7.1)$$

has two obvious integrals, H and the total momentum $\sum_n p_n$, and no other evident continuous symmetry: apart from a common translation, no one-parameter family of transformations of the positions leaves H invariant. (The reflection $q_n \mapsto -q_{N+1-n}$ does, but it is discrete and gives no integral.) In the early 1970s Ford's numerical experiments on the periodic chain suggested that it was nevertheless integrable, and in 1974 Hénon and Flaschka independently proved it (Toda, §2.10). Flaschka's construction, done here for the open chain, is the subject of § 7.6. It works by finding a matrix L , built from the q 's and p 's, whose eigenvalues do not move. N eigenvalues, N integrals.

Two warnings before we start. The method manufactures *conserved* quantities; whether they are independent and in involution is a separate question (§ 7.3). And there is no algorithm that finds L for a given H : Audin calls the search “cooking”.¹ What §§ 7.4 and 7.6 show is that, with the right ansatz, the cooking can be a derivation rather than a guess.

¹M. Audin, *Spinning Tops*, Introduction, §4.2, p. 13: “how to find a Lax form for a given system. I know no satisfying answer to this question.”

7.2 The Lax equation and isospectrality

Definition 7.1 (Lax pair). Let H be a Hamiltonian on a phase space $\mathcal{M}$. A Lax pair for H is a pair of smooth functions $L, M: \mathcal{M} \rightarrow \text{Mat}_N(\mathbb{C})$ such that along every solution $x(t)$ of Hamilton's equations, $L(t) = L(x(t))$ and $M(t) = M(x(t))$ satisfy the Lax equation

$$\frac{dL}{dt} = [M, L] = ML - LM. \quad (7.2)$$

L is the Lax matrix; N is its size, unrelated to $\dim \mathcal{M}$.

The definition asks that Hamilton's equations *imply* (7.2). In good cases the converse holds too, and (7.2) is the equations of motion, written as a matrix identity; we will see this for the oscillator, and for Toda up to the motion of the centre of mass, which the Lax matrix cannot see.

Theorem 7.2 (Isospectrality). Let (L, M) be a Lax pair for H and $x(t)$ a solution of Hamilton's equations defined on an interval containing 0. Then:

- (i) $\text{tr } L^k$ is constant in t for every $k \geq 1$;
- (ii) $L(t) = U(t) L(0) U(t)^{-1}$, where U solves $\dot{U} = M(t)U$, $U(0) = \mathbb{1}$, and is invertible;
- (iii) the characteristic polynomial $\det(\eta - L(t))$ is independent of t : the eigenvalues of L , with their multiplicities, are conserved.

Consequently each $I_k = \text{tr } L^k$ is a first integral, $\{I_k, H\} = 0$ (Definition 1.13); if L is complex, the real and imaginary parts of each I_k are. If M is antisymmetric, $U(t)$ is orthogonal; if M is anti-Hermitian, unitary.

Proof. (i) By the product rule and (7.2), and then because the trace is invariant under cyclic permutation,

$$\frac{d}{dt} \text{tr } L^k = k \text{tr}(L^{k-1} \dot{L}) = k \text{tr}(L^{k-1} ML - L^k M) = 0.$$

(The product rule gives k terms $\text{tr}(L^j \dot{L} L^{k-1-j})$; cyclicity makes them equal.)

(ii) The equation $\dot{U} = M(t)U$ is linear with coefficients continuous in t , so its solution exists on the whole interval.² Let V solve $\dot{V} = -V M(t)$, $V(0) = \mathbb{1}$. Then $\frac{d}{dt}(VU) = -VMU + VMU = 0$, so $VU = \mathbb{1}$ and U is invertible with $U^{-1} = V$. Now

$$\frac{d}{dt}(VLU) = -VMLU + V(ML - LM)U + VL MU = 0,$$

so $U^{-1}L(t)U = L(0)$.

- (iii) $\det(\eta - ULU^{-1}) = \det(U(\eta - L)U^{-1}) = \det(\eta - L)$.

Every point of $\mathcal{M}$ lies on some solution, and $dI_k/dt = \{I_k, H\}$ (Definition 1.13), so (i) gives $\{I_k, H\} = 0$ everywhere. Finally, if $M^T = -M$ then $\frac{d}{dt}(U^T U) = U^T(M^T + M)U = 0$, and the same line with the conjugate transpose U^* gives the unitary case. ■

Part (ii) says more than (i): the flow moves L inside one conjugacy class. For L real symmetric and M antisymmetric, $[M, L]$ is symmetric, so $L(t)$ stays symmetric and moves by orthogonal conjugation (Figure 7.1).

For an $N \times N$ matrix only N of the $\text{tr } L^k$ can be independent: by the Cayley–Hamilton theorem L^N is a combination of $\mathbb{1}, L, \dots, L^{N-1}$ with coefficients that are polynomials in

²A linear equation cannot blow up in finite time: Grönwall's inequality bounds $\|U(t)\|$ by $e^{\int \|M\|}$. This is the only fact about ODEs used beyond Theorem 1.5.

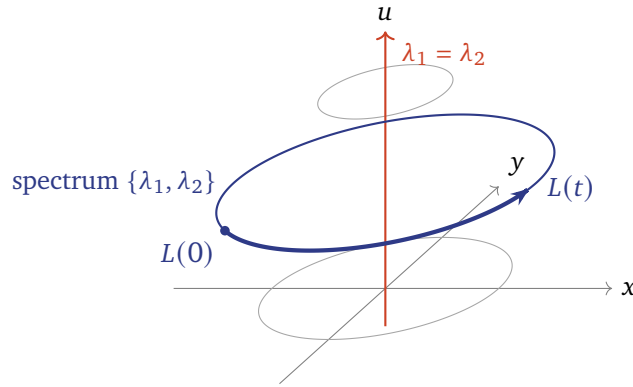


Figure 7.1: Isospectral flow in the smallest case. A real symmetric 2×2 matrix $\begin{pmatrix} u+x & y \\ y & u-x \end{pmatrix}$ has eigenvalues $u \pm \sqrt{x^2 + y^2}$, so fixing the spectrum fixes the height u and the radius $\sqrt{x^2 + y^2}$: the matrices with a given spectrum $\lambda_1 \neq \lambda_2$ form one horizontal circle. A Lax flow with antisymmetric M cannot leave its circle. On the vertical axis the two eigenvalues coincide and the “circle” is a point.

$\text{tr } L, \dots, \text{tr } L^N$, so every $\text{tr } L^k$ with $k > N$ is a polynomial in the first N . Equivalently, the conserved data is the characteristic polynomial, which has N coefficients.

7.3 What a Lax pair does not give: the IOU

[Theorem 7.2](#) produces first integrals. [Definition 4.6](#) needs n of them, independent and in involution. The next proposition, which this lecture will keep returning to, says that these are three different things.

Proposition 7.3 (What isospectrality does not give).

- (a) A Lax pair exists for every Hamiltonian. $L = (H)$, a 1×1 matrix, and $M = (0)$ satisfy (7.2).
- (b) Conservation is not involution. For a particle in a central potential on $\mathbb{R}^3$, $L = \text{diag}(\ell_1, \ell_2)$ and $M = 0$ form a Lax pair, and

$$\{\text{tr } L, \text{tr } L^2\} = \{\ell_1 + \ell_2, \ell_1^2 + \ell_2^2\} = 2\ell_3(\ell_2 - \ell_1) \neq 0.$$

- (c) The spectrum can miss an integral. The Lax matrix of the free rigid body in § 7.5 has only one independent spectral invariant, $|\ell|^2$; unless the three principal moments are equal, the energy, also conserved, is not a function of its spectrum.

Proof. (a) Both sides of (7.2) vanish. (b) The ℓ_i are first integrals ([Example 4.4](#)), so $\dot{L} = 0 = [0, L]$; the bracket follows from $\{\ell_1, \ell_2\} = \ell_3$ (4.3) and the Leibniz rule: $\{\ell_1, \ell_2^2\} + \{\ell_2, \ell_1^2\} = 2\ell_2\ell_3 - 2\ell_1\ell_3$. (c) is (7.7) below. ■

The IOU. Having a Lax pair is no property at all (a). What matters is the spectral invariants it delivers, and nothing in (7.2) makes them commute (b) or makes enough of them independent (c). [Lecture 8](#) supplies the missing input, a condition on the Poisson brackets of the entries of L , and proves involution from it. Independence is checked model by model; [Lecture 9](#) does it for Toda.

The Lax equation is a statement about how L moves along *one* flow, the flow of H .

Involution is a statement about the brackets of the I_k with each other, which no single flow can see. That is why the debt cannot be paid inside this lecture.

7.4 The harmonic oscillator, derived

Take $H = \frac{1}{2}(p^2 + q^2)$, so $\dot{q} = p$, $\dot{p} = -q$. We look for a Lax pair in two steps, the same two steps that § 7.6 will take for Toda.

Step 1: choose L . We want H among the spectral invariants. A real symmetric traceless 2×2 matrix $\begin{pmatrix} x & y \\ y & -x \end{pmatrix}$ squares to $(x^2 + y^2)\mathbb{1}$, a sum of two squares, and $2H$ is a sum of two squares. So take $x = p$, $y = q$:

$$L = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}, \quad L^2 = 2H \mathbb{1}, \quad \text{tr } L^2 = 4H. \quad (7.3)$$

Its eigenvalues are $\pm\sqrt{2H}$.

Step 2: solve for M . Hamilton's equations give $\dot{L} = \begin{pmatrix} -q & p \\ p & q \end{pmatrix}$. The unknown M enters $[M, L] = \dot{L}$ linearly, so this is a linear system for four unknowns. It never has a unique solution: if M works, so does $M + D$ for any D commuting with L , and here $D = \alpha\mathbb{1} + \beta L$ commutes with L . Uniqueness has to be imposed. Ask for M antisymmetric, so that the flow is an orthogonal conjugation and keeps L symmetric (Theorem 7.2). With $M = \begin{pmatrix} 0 & -m \\ m & 0 \end{pmatrix}$,

$$[M, L] = 2m \begin{pmatrix} -q & p \\ p & q \end{pmatrix},$$

so $m = \frac{1}{2}$ wherever $(q, p) \neq 0$, hence everywhere by continuity:

$$M = \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (7.4)$$

Here the Lax pair also solves the equations. M is constant, so $U(t) = e^{tM}$ is the rotation through $t/2$, and conjugating L by a rotation through θ rotates the vector (p, q) through 2θ . Theorem 7.2(ii) therefore says that (p, q) rotates through t , which is the oscillator's motion. The eigenvalue $\sqrt{2H}$ is the radius of the circle; with one degree of freedom it is the only integral needed. Lecture 8 uses exactly this L as its first worked example.

7.5 The Euler top

A rigid body turning freely about its centre of mass, described in the *body frame*, the principal axes of inertia moving with the body. The principal moments are $I_1, I_2, I_3 > 0$; the angular momentum and the angular velocity have body components ℓ and ω , with $\ell_i = I_i \omega_i$. In space ℓ is constant; the body frame turns with angular velocity ω , so $0 = \dot{\ell} + \omega \times \ell$, which are *Euler's equations*

$$\dot{\ell} = \ell \times \omega, \quad \omega_i = \ell_i / I_i, \quad (7.5)$$

in components $I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3$ and cyclically. We use them only as a differential equation on $\mathbb{R}^3$. (These I_i are moments of inertia, not the integrals I_k of Theorem 7.2 nor Lecture 5's actions; the standard names are kept.) (The phase space of the body is $T^*\text{SO}(3)$, of dimension 6 (Example 4.7); the three components of ℓ are functions on it, and (7.5) is the part of the motion that does not involve the orientation.)

The Lax form

For $a \in \mathbb{R}^3$ let $\hat{a}$ be the antisymmetric matrix with $\hat{a}v = a \times v$:

$$\hat{a} = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}.$$

Then $[\hat{a}, \hat{b}] = \widehat{a \times b}$: apply both sides to v and use the Jacobi identity of the cross product, $a \times (b \times v) - b \times (a \times v) = (a \times b) \times v$. Hence, with $L = \hat{\ell}$ and $\Omega = \hat{\omega}$, Euler's equations are

$$\frac{dL}{dt} = \widehat{\ell \times \omega} = [L, \Omega], \quad (7.6)$$

a Lax equation with $M = -\Omega$. The student already knew it; only the notation is new.

What does it conserve? L is real antisymmetric, so $\text{tr } L = \text{tr } L^3 = 0$ and $\text{tr } L^2 = -2|\ell|^2$; indeed

$$\det(\eta - L) = \eta^3 + |\ell|^2 \eta, \quad \text{eigenvalues } 0, \pm i|\ell|. \quad (7.7)$$

The spectrum knows $|\ell|^2$ and nothing else. But the energy

$$H = \sum_i \frac{\ell_i^2}{2I_i}$$

is also conserved, since $\dot{H} = \omega \cdot \dot{\ell} = \omega \cdot (\ell \times \omega) = 0$, and unless $I_1 = I_2 = I_3$ it is not a function of $|\ell|^2$. This is [Proposition 7.3\(c\)](#): a genuine Lax pair, for a genuinely integrable system, whose spectrum misses the second integral. [§ 7.9](#) gets it back.

The motion, from the two integrals

Take $I_1 < I_2 < I_3$. Both integrals together confine $\ell(t)$ to the intersection of the sphere $|\ell| = \ell_0 > 0$ with the ellipsoid $\sum \ell_i^2/I_i = 2H$ ([Figure 7.2](#)). On the sphere H is smallest at $\pm \ell_0 e_3$ and largest at $\pm \ell_0 e_1$, and near these four points the level curves are small closed loops. The equilibria of (7.5), where $\ell \times \omega = 0$, are the six points $\pm \ell_0 e_i$: steady rotation about a principal axis. The middle level, $2H = \ell_0^2/I_2$, is

$$\ell_1^2 \left(\frac{1}{I_1} - \frac{1}{I_2} \right) = \ell_3^2 \left(\frac{1}{I_2} - \frac{1}{I_3} \right), \quad (7.8)$$

two planes $\ell_3 = \pm k\ell_1$ through the e_2 axis, cutting the sphere in two great circles that cross at $\pm \ell_0 e_2$: the *separatrix*.

Theorem 7.4 (The tennis-racket theorem). *Let $I_1 < I_2 < I_3$ and $\ell_0 > 0$. Steady rotation about the axis of smallest or of largest moment is stable: for every $\varepsilon > 0$ there is $\delta > 0$ such that $|\ell(0) - \ell_0 e_1| < \delta$ implies $|\ell(t) - \ell_0 e_1| < \varepsilon$ for all t , and the same for e_3 . Steady rotation about the middle axis is unstable: there are solutions starting arbitrarily close to $\ell_0 e_2$ that tend to $-\ell_0 e_2$.*

At the board the proof is [Figure 7.2](#): near $\pm e_1$ and $\pm e_3$ the conserved quantities confine ℓ to small loops, while every neighbourhood of $\pm e_2$ meets the separatrix, which runs to the opposite pole. The reading turns both halves into estimates.

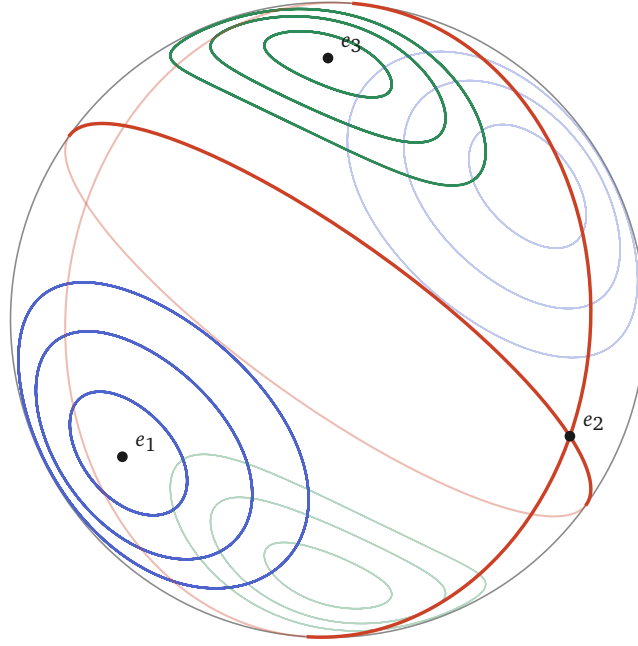


Figure 7.2: The free Euler top in the body frame, $I = (1, 2, 3)$, $|\ell| = 1$. The closed loops round the axes of smallest and largest moment, $\pm e_1$ (blue) and $\pm e_3$ (green), are orbits of (7.5), computed by integrating it (`scripts/figures/euler_sphere.py`, which checks that $|\ell|$ and H are conserved along each). The separatrix (red) is drawn from (7.8): two great circles, made of the equilibria $\pm e_2$ and four orbits running between them. The far hemisphere is faint.

Proof (★). Stability at e_1 . The function

$$G = \frac{|\ell|^2}{I_1} - 2H = \ell_2^2 \left(\frac{1}{I_1} - \frac{1}{I_2} \right) + \ell_3^2 \left(\frac{1}{I_1} - \frac{1}{I_3} \right) \geq c (\ell_2^2 + \ell_3^2), \quad c = \frac{1}{I_1} - \frac{1}{I_2} > 0,$$

is conserved and vanishes at $\ell_0 e_1$. So $\ell_2^2 + \ell_3^2 \leq G(0)/c$ for all t , and $G(0)$ is small when $\ell(0)$ is near $\ell_0 e_1$. Since $|\ell|$ is conserved, $\ell_1^2 = |\ell|^2 - \ell_2^2 - \ell_3^2$ stays close to ℓ_0^2 ; ℓ_1 cannot pass through 0, so it keeps its sign and stays close to ℓ_0 . At e_3 use $2H - |\ell|^2/I_3$ instead.

Instability at e_2 . The components of $\ell \times \omega$ are $\ell_2 \ell_3 (1/I_3 - 1/I_2)$, $\ell_3 \ell_1 (1/I_1 - 1/I_3)$ and $\ell_1 \ell_2 (1/I_2 - 1/I_1)$; with distinct moments they all vanish only when two components of ℓ do, so the equilibria are the points $\pm \ell_0 e_i$. Each open arc of the separatrix between $\ell_0 e_2$ and $-\ell_0 e_2$ contains no equilibrium, so a solution started on it moves along it with nowhere-vanishing speed, so monotonically; being bounded, it converges as $t \rightarrow \pm\infty$, and the limits can only be equilibria. So each arc runs from one end to the other. It remains to see that some arc leaves $\ell_0 e_2$. The map $(\ell_1, \ell_2, \ell_3) \mapsto (-\ell_1, \ell_2, \ell_3)$ combined with $t \mapsto -t$ takes solutions of (7.5) to solutions (check it on the three components, each of which is a product of the other two ℓ 's), fixes $\ell_0 e_2$, exchanges the two planes of (7.8) and reverses time. So if the arc in one plane runs into $\ell_0 e_2$, its mirror image runs out of it, towards $-\ell_0 e_2$. Its points can be taken as close to $\ell_0 e_2$ as we like. ■

Throw a tennis racket or a phone spinning about its middle axis: it flips, and off the separatrix it flips back periodically, because a nearby orbit follows a loop that runs close to the separatrix, round to near $-\ell_0 e_2$ and back (Figure 7.3).

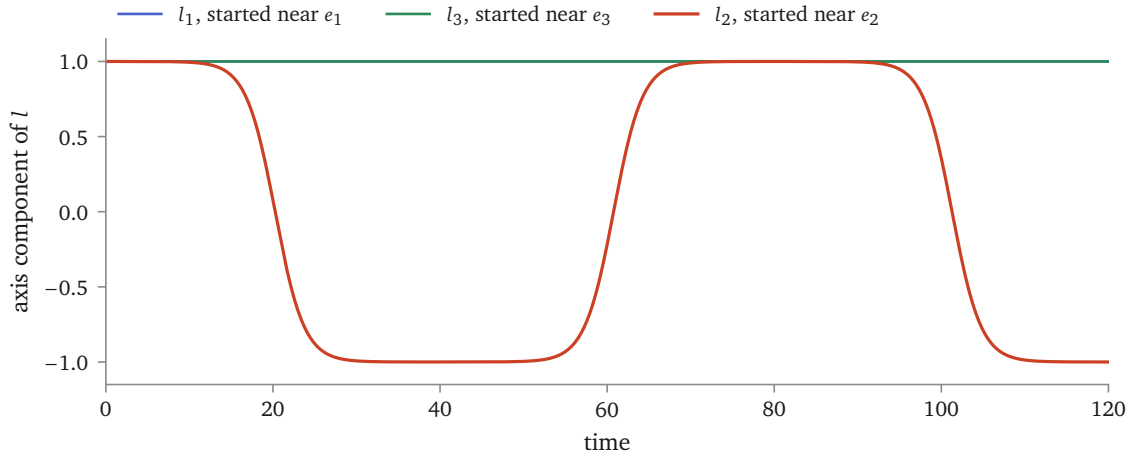


Figure 7.3: The tennis-racket theorem, numerically: $I = (1, 2, 3)$, $|\ell| = 1$, each run started 0.01 radians from a principal axis. The axis components ℓ_1 (started near e_1) and ℓ_3 (near e_3) stay at 1 to within 2×10^{-4} and their curves lie on top of each other; ℓ_2 (near e_2) flips between +1 and -1 over and over (scripts/figures/tennis_racket.py, which asserts both).

Reading

Poinsot's construction

The same two integrals give Poinsot's picture of the motion *in space*. In the body, the angular velocity lies on the *energy ellipsoid* $E = \{v : v \cdot Iv = 2H\}$, since $\omega \cdot I\omega = \omega \cdot \ell = 2H$. The normal to E at ω is the gradient $2I\omega = 2\ell$, and the tangent plane there lies at distance $\omega \cdot \ell / |\ell| = 2H / |\ell|$ from the centre. In space ℓ is a fixed vector and $2H / |\ell|$ a fixed number, so this tangent plane is *the same plane at all times*, the *invariable plane*. The point of contact is ω , which lies on the instantaneous axis of rotation and so has zero velocity. Hence:

Proposition 7.5 (Poinsot). *The energy ellipsoid, carried by the body, rolls without slipping on a fixed plane perpendicular to ℓ at distance $2H / |\ell|$ from the centre.*

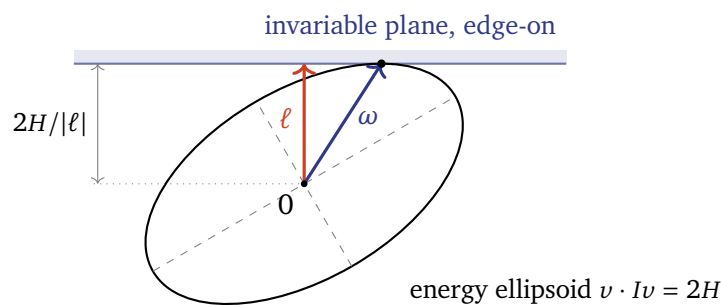


Figure 7.4: Poinsot's construction. ℓ is fixed in space; the energy ellipsoid, fixed in the body, touches the invariable plane at ω , the one point of the ellipsoid with zero velocity, and so rolls on it without slipping. Drawn in the plane containing ℓ and ω .

Remark 7.6 (The other tops). The symmetric heavy top of Lagrange and the Kowalevski top also have Lax pairs, both with a spectral parameter of the kind § 7.9 introduces · cited: Audin, *Spinning Tops*, ch. II §2 and ch. III §2. We do not use them.

7.6 The open Toda chain: Flaschka variables, L and M derived

N unit masses on a line, positions $q_1, \dots, q_N$, each repelled by its neighbours through the potential $e^{q_n - q_{n+1}}$, which is large when particle n has overtaken particle $n + 1$ and small when they are far apart. With H as in (7.1), Hamilton's equations are

$$\dot{q}_n = p_n, \quad \dot{p}_n = e^{q_{n-1} - q_n} - e^{q_n - q_{n+1}}, \quad (7.9)$$

where the terms involving q_0 and q_{N+1} are absent: the chain is *open*, and its end particles have one neighbour each. Lecture 9 says where this potential came from. Here we find its Lax pair in the same two steps as the oscillator's, after a change of variables that makes the equations polynomial.

Flaschka's variables

The exponentials are the obstacle. Set

$$a_n = e^{(q_n - q_{n+1})/2} \quad (n = 1, \dots, N-1), \quad b_n = p_n \quad (n = 1, \dots, N). \quad (7.10)$$

Then $a_n^2 = e^{q_n - q_{n+1}}$, and (7.9) becomes

$$\dot{a}_n = \frac{1}{2} a_n (b_n - b_{n+1}), \quad \dot{b}_n = a_{n-1}^2 - a_n^2, \quad (7.11)$$

with the convention $a_0 = a_N = 0$ — quadratic, with no exponential left. The map $(q, p) \mapsto (a, b)$ forgets one thing: shifting every q_n by the same constant changes no a_n . The centre of mass position is invisible to L ; it is recovered from $\dot{q}_n = p_n$ once the rest is known. Note also $a_n > 0$.

Convention. In this normalisation, frozen for the course, the eigenvalues of L have the units of momentum; Toda's book and Babelon–Bernard–Talon scale differently (see the references).

Step 1: L

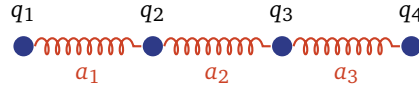
In the new variables

$$2H = \sum_{n=1}^N b_n^2 + 2 \sum_{n=1}^{N-1} a_n^2,$$

a sum of squares, each a_n^2 counted twice. For a real symmetric matrix, $\text{tr } L^2 = \sum_{j,k} L_{jk}^2$ is also a sum of squares, each off-diagonal entry counted twice because it appears as L_{jk} and L_{kj} . So the simplest L with $\text{tr } L^2 = 2H$ has each b_n alone in a diagonal slot and each a_n alone in a symmetric pair of off-diagonal slots. Which pair is a choice, and locality makes it: a_n is the bond between particles n and $n + 1$, so it goes in the slots $(n, n + 1)$ and $(n + 1, n)$. The result is the symmetric tridiagonal (Jacobi) matrix of Figure 7.5; for $N = 3$,

$$L = \begin{pmatrix} b_1 & a_1 & 0 \\ a_1 & b_2 & a_2 \\ 0 & a_2 & b_3 \end{pmatrix}, \quad \text{tr } L = \sum_n p_n, \quad \text{tr } L^2 = 2H. \quad (7.12)$$

Step 1 is a well-motivated choice, not a theorem; Step 2 is where it is tested.



$$L = \begin{array}{|c|c|c|c|} \hline b_1 & a_1 & 0 & 0 \\ \hline a_1 & b_2 & a_2 & 0 \\ \hline 0 & a_2 & b_3 & a_3 \\ \hline 0 & 0 & a_3 & b_4 \\ \hline \end{array}$$

bond n (red) fills the two slots $(n, n+1)$ and $(n+1, n)$; particle n 's momentum (indigo) fills (n, n) ; everything else is zero

Figure 7.5: Where the entries of Toda's L come from, for $N = 4$. The chain's nearest-neighbour structure becomes the tridiagonal band.

Step 2: M , from an ansatz

We want M with $[M, L] = \dot{L}$, where $\dot{L}$ is given by (7.11). As for the oscillator, take M antisymmetric, so that L stays symmetric. $\dot{L}$ is tridiagonal, and the commutator of two tridiagonal matrices has bandwidth at most two, so try the least that could work: M antisymmetric and tridiagonal,

$$M_{n,n+1} = -m_n, \quad M_{n+1,n} = m_n,$$

with $m_1, \dots, m_{N-1}$ unknown functions. Compare the three bands of $[M, L]$ with those of $\dot{L}$.

Second off-diagonal. $\dot{L}$ has none, but

$$[M, L]_{n,n+2} = M_{n,n+1}L_{n+1,n+2} - L_{n,n+1}M_{n+1,n+2} = -(m_n a_{n+1} - a_n m_{n+1})$$

must vanish. So $m_n/a_n = m_{n+1}/a_{n+1}$ for every n : $m_n = c a_n$ with one common factor c .

First off-diagonal. Only two terms survive:

$$[M, L]_{n,n+1} = M_{n,n+1}L_{n+1,n+1} - L_{n,n}M_{n,n+1} = m_n(b_n - b_{n+1}),$$

and (7.11) asks for $\frac{1}{2}a_n(b_n - b_{n+1})$. As this must hold for all b , $m_n = \frac{1}{2}a_n$: $c = \frac{1}{2}$, determined.

Diagonal. Nothing is left to choose, so this is the test:

$$[M, L]_{nn} = M_{n,n+1}a_n + M_{n,n-1}a_{n-1} - a_n M_{n+1,n} - a_{n-1} M_{n-1,n} = 2(m_{n-1}a_{n-1} - m_n a_n) = a_{n-1}^2 - a_n^2,$$

which is $\dot{b}_n$. It could have failed; it does not.

Proposition 7.7 (The Toda Lax pair). With L the symmetric tridiagonal matrix with diagonal b_n and off-diagonal a_n , and M the antisymmetric tridiagonal matrix with

$$M_{n,n+1} = -\frac{1}{2}a_n, \quad M_{n+1,n} = \frac{1}{2}a_n, \quad (7.13)$$

the Toda equations (7.11) are equivalent to $\dot{L} = [M, L]$. Among antisymmetric matrices, M is the only solution.

Proof. The three band computations above prove the Lax equation, and show that the tridiagonal ansatz has exactly this solution. The equivalence holds because the bands of $\dot{L} = [M, L]$ are (7.11), entry by entry. For uniqueness among all antisymmetric matrices: if

M' also works, $D = M - M'$ is antisymmetric and commutes with L . A Jacobi matrix with all $a_n \neq 0$ has N distinct eigenvalues: if $Lv = \eta v$, row 1 gives v_2 from v_1 , row 2 gives v_3 , and so on, so each eigenspace is at most one-dimensional. Diagonalise $L = O\Lambda O^T$ with O orthogonal; then $O^T D O$ commutes with the diagonal Λ of distinct entries, so it is diagonal, and it is antisymmetric, so it is zero. ■

So, to “where did L come from?”: from $\text{tr } L^2 = 2H$ and locality. And “where did M come from?”: it was solved for, and it was the only possibility.

The integrals

By [Theorem 7.2](#), the N functions $\text{tr } L^k$, $k = 1, \dots, N$, are first integrals. The first three:

$$\text{tr } L = \sum_n p_n, \quad \text{tr } L^2 = 2H, \quad \text{tr } L^3 = \sum_n p_n^3 + 3 \sum_{n=1}^{N-1} e^{q_n - q_{n+1}} (p_n + p_{n+1}). \quad (7.14)$$

Total momentum, energy, and an integral cubic in the momenta. By [Lecture 1](#)’s one-line Noether theorem it too generates a symmetry of H , but not a transformation of the positions alone, since its flow moves each q_n at a rate $\partial(\text{tr } L^3)/\partial p_n$ that depends on the momenta. That is why nobody saw it. [Figure 7.6](#) watches them at work for $N = 3$: the momenta change completely during a collision while the three eigenvalues of L do not change at all. What the eigenvalues *mean* is [Lecture 9](#)’s business, and the figure already hints at it.

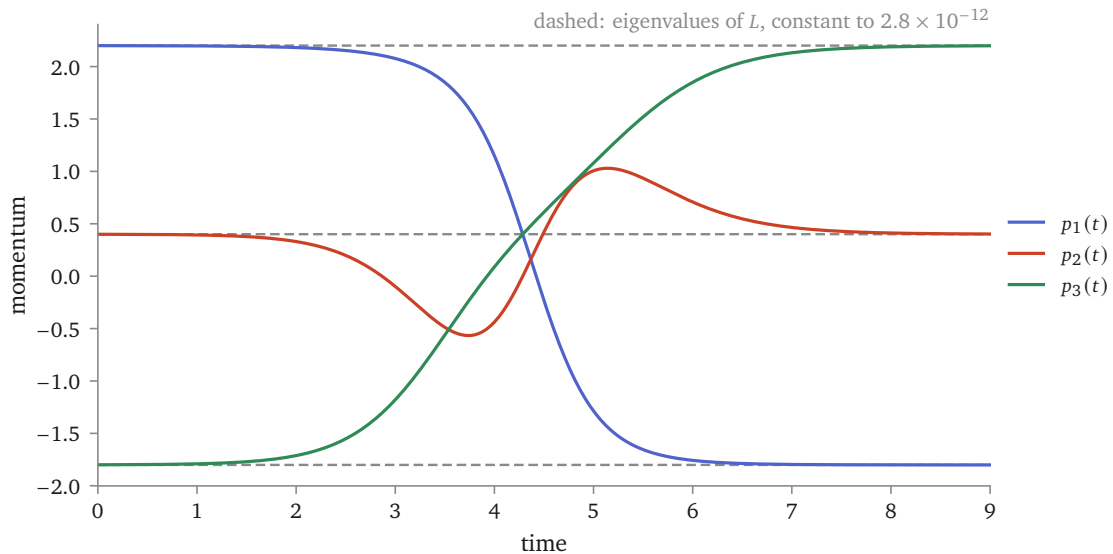


Figure 7.6: The open Toda chain, $N = 3$: momenta $p_n(t)$ (solid) against the eigenvalues of L (dashed grey), computed from (7.10) at every time step. The eigenvalues are constant to the stated precision while the momenta are rearranged (scripts/figures/toda_isospectral.py, which also checks $\text{tr } L^2 = 2H$ and energy conservation).

Whether these N integrals are independent and in involution is exactly [Proposition 7.3](#)’s question. Both are true, and both need more than this lecture: involution is [Lecture 8](#)’s theorem applied in [Lecture 9](#), and independence is proved in [Lecture 9](#).

7.7 Calogero–Moser: Moser’s matrix, stated

N particles on a line with a repulsive inverse-square pair potential,

$$H = \frac{1}{2} \sum_{j=1}^N p_j^2 + \sum_{j < k} \frac{g^2}{(q_j - q_k)^2}, \quad g > 0. \quad (7.15)$$

Unlike Toda, every particle interacts with every other, and the Lax matrix Moser found is full.

Proposition 7.8 (Moser’s Lax pair · cited). *On the region where the q_j are distinct, let*

$$L_{jk} = p_j \delta_{jk} + \frac{ig(1 - \delta_{jk})}{q_j - q_k}, \quad M_{jk} = \frac{ig(1 - \delta_{jk})}{(q_j - q_k)^2} - ig \delta_{jk} \sum_{l \neq j} \frac{1}{(q_j - q_l)^2}. \quad (7.16)$$

Then $\dot{L} = [M, L]$ along the flow of (7.15). L is Hermitian, M anti-Hermitian, and $\text{tr } L^2 = 2H$.

The proof is a direct computation, which we do not do here: [Lecture 11](#) derives the pair by solving for M , and [Lecture 12](#) explains where it comes from. (It is checked symbolically for $N = 2, 3$ in `scripts/check-17-algebra.py`.) The check $\text{tr } L^2 = 2H$ is one line: the off-diagonal part contributes $\sum_{j \neq k} \frac{ig}{q_j - q_k} \cdot \frac{ig}{q_k - q_j} = \sum_{j \neq k} \frac{g^2}{(q_j - q_k)^2}$, twice the potential.

L is Hermitian and M anti-Hermitian, so $L(t)$ moves by unitary conjugation ([Theorem 7.2](#)) and its N eigenvalues are real first integrals.

7.8 Gauge freedom

A Lax pair is not unique, and the non-uniqueness has a structure worth knowing.

Proposition 7.9 (Gauge transformations). *Let (L, M) satisfy (7.2) along a solution, and let $g(t)$ be invertible and differentiable in t (for instance a function on phase space, evaluated along the solution). Then*

$$\tilde{L} = gLg^{-1}, \quad \tilde{M} = gMg^{-1} + \dot{g}g^{-1} \quad (7.17)$$

satisfy $\dot{\tilde{L}} = [\tilde{M}, \tilde{L}]$, and $\tilde{L}$ has the same spectrum as L .

Proof. With $\frac{d}{dt}g^{-1} = -g^{-1}\dot{g}g^{-1}$,

$$\dot{\tilde{L}} = \dot{g}Lg^{-1} + g[M, L]g^{-1} - gLg^{-1}\dot{g}g^{-1} = [\dot{g}g^{-1}, \tilde{L}] + [gMg^{-1}, \tilde{L}]. \quad \blacksquare$$

Everything spectral is gauge invariant. Everything else is not: symmetry, the band structure, M itself. For Toda, conjugate by $D = \text{diag}(d_n)$ with $d_1 = 1$, $d_{n+1} = a_n d_n$. Then DLD^{-1} has b_n on the diagonal, 1 above it and $a_n^2 = e^{q_n - q_{n+1}}$ below it: not symmetric, but with entries that are simpler functions of (q, p) , and with the same characteristic polynomial. [Figure 7.7](#) shows the two as different paths on one isospectral set.

The warning to carry forward: the Poisson brackets of the *entries* of L , which [Lecture 8](#) will make the key to involution, change under a gauge transformation even though the spectrum does not. And in [Lecture 10](#) a gauge transformation is exactly the step that makes the periodic chain’s Lax representation *local*, built site by site.

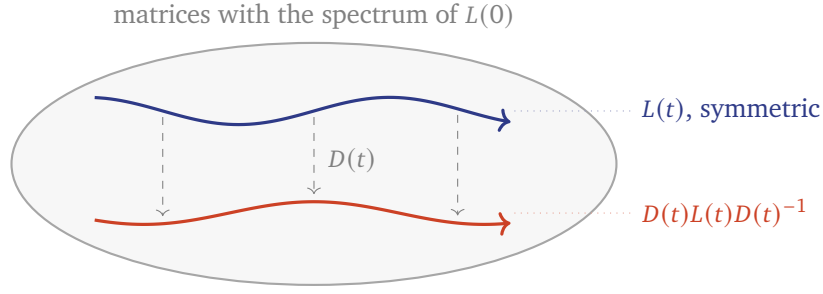


Figure 7.7: Gauge freedom, schematically. Both curves lie in the set of matrices with a given spectrum; each is the image of the same Toda trajectory under a different Lax matrix, and the gauge transformation $D(t)$ (dashed) carries one to the other at each instant.

7.9 The spectral parameter

The Euler top left a problem: for a body whose moments are not all equal, its Lax matrix $\hat{\ell}$ conserves $|\ell|^2$ and misses H (Proposition 7.3(c)). Manakov's repair (1976) is to let the Lax matrix depend on an extra complex number λ , the *spectral parameter*.

Put $A = \text{diag}(A_1, A_2, A_3)$ with $A_i = \frac{1}{2}(I_1 + I_2 + I_3) - I_i$, so that $A_2 + A_3 = I_1$ and cyclically. Since $(A\Omega + \Omega A)_{jk} = (A_j + A_k)\Omega_{jk}$, this gives $\hat{\ell} = A\Omega + \Omega A$. Now set

$$L(\lambda) = \hat{\ell} + \lambda A^2, \quad M(\lambda) = -(\Omega + \lambda A). \quad (7.18)$$

Then $\dot{L}(\lambda) = [M(\lambda), L(\lambda)]$ for every λ : the λ^0 term is (7.6), the λ^2 term is $[A, A^2] = 0$, and the λ^1 term is $-\left[\Omega, A^2\right] - [A, A\Omega + \Omega A]$, which vanishes because $[A, A\Omega + \Omega A] = A^2\Omega - \Omega A^2$.

So Theorem 7.2 applies at each λ separately, and the characteristic polynomial of $L(\lambda)$ is conserved as a polynomial in two variables. It is

$$\det(\eta - L(\lambda)) = \det(\eta - \lambda A^2) + |\ell|^2 \eta - \lambda \sum_i A_i^2 \ell_i^2, \quad (7.19)$$

where the first term involves only the constants A_i , and

$$\sum_i A_i^2 \ell_i^2 = 2I_1 I_2 I_3 H + \alpha |\ell|^2, \quad \alpha = \frac{1}{4}(I_1^2 + I_2^2 + I_3^2 - 2I_1 I_2 - 2I_2 I_3 - 2I_3 I_1).$$

The coefficient of $\lambda \eta^0$ contains H : the integral that $\hat{\ell}$ missed is a spectral invariant of $L(\lambda)$. (For a spherical body A is a multiple of $\mathbb{1}$, λA^2 only shifts the spectrum, and there was nothing to recover.) All three identities are checked symbolically in `scripts/check-17-algebra.py`.

This is the first thing a spectral parameter is for: *one matrix family carries more invariants than one matrix*, since every coefficient of every power of λ in $\text{tr } L(\lambda)^k$ is conserved. There are two more, and they are the reason Part I keeps λ from here on.

- *A curve.* The conserved polynomial $\det(\eta - L(\lambda))$ defines the *spectral curve* $\Gamma = \{(\lambda, \eta) \in \mathbb{C}^2 : \det(\eta - L(\lambda)) = 0\}$, which the flow leaves unchanged; its coefficients are the integrals. Lecture 10 builds it for the periodic Toda chain, whose Lax matrix carries in its corners the Floquet multiplier of Bloch's theorem, and reads the bands, the gaps and the action variables off it. (Here η is local notation; Lecture 10 names the two variables differently and says so.)
- *A bracket.* The Sklyanin bracket of Lecture 8 compares $L(\lambda)$ with $L(\mu)$ at two different values of the parameter, through $r(\lambda - \mu)$. That comparison is what forces $\{\text{tr } L(\lambda)^n, \text{tr } L(\mu)^m\} = 0$, and so pays this lecture's IOU.

Summary.

- *Lax pair* (Definition 7.1): $\dot{L} = [M, L]$ along the flow. *Isospectrality* (Theorem 7.2): $L(t) = UL(0)U^{-1}$, so the spectrum and every $\text{tr } L^k$ are conserved; at most N are independent.
- *The IOU* (Proposition 7.3): a Lax pair always exists, its invariants need not commute, and they can miss an integral. Involution is Lecture 8's; independence is model by model.
- *The method*: choose L with $\text{tr } L^2 \propto H$, then solve the linear equation $[M, L] = \dot{L}$ for M within an ansatz; M is only ever unique up to the commutant of L . Oscillator: (7.3)–(7.4).
- *Euler top*: $\hat{\ell} = [\hat{\ell}, \hat{\omega}]$; the spectrum sees only $|\ell|^2$. The two integrals give the tennis-racket theorem (Theorem 7.4) and, in the reading, Poincaré's rolling ellipsoid.
- *Open Toda* (Proposition 7.7): Flaschka's variables (7.10), L symmetric tridiagonal with $\text{tr } L^2 = 2H$, M derived and unique among antisymmetric matrices; integrals (7.14).
- *Calogero–Moser* (Proposition 7.8): Moser's L , stated; Lecture 11 derives it.
- *Gauge* (Proposition 7.9) changes L but not its spectrum. *Spectral parameter* (7.18): recovers the Euler top's energy, defines the spectral curve (Lecture 10) and carries the Sklyanin bracket (Lecture 8).

Tutorial. Exercise Sheet 7.

References

Toda, *Theory of Nonlinear Lattices*, §2.10 (Hénon and Flaschka) and §3.1 (the matrix formalism); H. Flaschka, Phys. Rev. B 9 (1974) 1924; Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, §4.6 (the open Toda Lax pair for any simple Lie algebra); Audin, *Spinning Tops*, Introduction (the spectral curve), ch. IV §1 (Euler and Manakov), ch. II §2 and ch. III §2 (Lagrange and Kowalevski); J. Moser, Adv. Math. 16 (1975) 197 (the Calogero–Moser Lax pair); S. V. Manakov, Funct. Anal. Appl. 10 (1976) 328. **Conventions.** Toda's book writes $a_n = \frac{1}{2}e^{(q_n - q_{n+1})/2}$, $b_n = \frac{1}{2}p_n$, so its L is half ours; Babelon–Bernard–Talon's q is half ours and their bracket is $\{p_i, q_j\} = \frac{1}{2}\delta_{ij}$.